

Math 432/532: Introduction to Topology II

Solutions to HW #6

1. Prove Lemma 8.8 from the book (also stated in class), which says that the map χ is a chain map.

Solution: We need to show that, for any oriented i -simplex σ of K , we have

$$\partial_i^{K'} \circ \chi_i(\sigma) = \chi_{i-1} \circ \partial_i^K(\sigma).$$

If σ does not have A as a face, then both sides are equal to $\partial_i^K(\sigma)$. On the other hand, suppose that $A = \Delta_{(v_0, \dots, v_k)}$ and $\sigma = (v_0, \dots, v_k, \dots, v_i)$. In this case,

$$\partial_i^K(\sigma) = \sum_{j=0}^i (-1)^k (v_0, \dots, \widehat{v_j}, \dots, v_i),$$

and therefore

$$\chi_{i-1} \circ \partial_i^K(\sigma) = \sum_{j=0}^k (-1)^k (v_0, \dots, \widehat{v_j}, \dots, v_i) + \sum_{j=k+1}^i (-1)^j \sum_{l=0}^k (-1)^l (v, v_0, \dots, \widehat{v_l}, \dots, \widehat{v_j}, \dots, v_i).$$

On the other hand, we have

$$\begin{aligned} \partial_i^{K'} \circ \chi_i(\sigma) &= \partial_i \sum_{j=0}^k (-1)^j (v, v_0, \dots, \widehat{v_j}, \dots, v_i) \\ &= \sum_{j=0}^k (-1)^j (v_0, \dots, \widehat{v_j}, \dots, v_i) \\ &\quad + \sum_{j=0}^k (-1)^j \sum_{l=1}^{j-1} (-1)^{l+1} (v, v_0, \dots, \widehat{v_l}, \dots, \widehat{v_j}, \dots, v_i) \\ &\quad + \sum_{j=0}^k (-1)^j \sum_{l=j+1}^k (-1)^l (v, v_0, \dots, \widehat{v_j}, \dots, \widehat{v_l}, \dots, v_i) \\ &\quad + \sum_{j=0}^k (-1)^j \sum_{l=k+1}^i (-1)^l (v, v_0, \dots, \widehat{v_j}, \dots, \widehat{v_l}, \dots, v_i). \end{aligned}$$

In this sum, the second and third lines cancel each other out, whereas the first and fourth lines match the two parts of our expression for $\chi_{i-1} \circ \partial_i^K(\sigma)$ (with the roles of j and l reversed in the second part).

2. Suppose that $s : |K^m| \rightarrow |L|$ is a simplicial approximation of $f : |K| \rightarrow |L|$, $n > m$, and $\theta : |K^n| \rightarrow |K^m|$ is a standard simplicial map (page 188). Show that $s \circ \theta : |K^n| \rightarrow |L|$ is also a simplicial approximation of f .

Solution: Recall that, for any point $y \in |L|$, the **carrier** $\Delta(y)$ of y is the smallest simplex in L containing y . Suppose that $x \in |K| = |K^m| = |K^n|$. Since s is a simplicial approximation of f , we know that $s(x) \in \Delta(f(x))$, the carrier of $f(x)$. Equivalently, s takes the simplex $\Delta^{K^m}(x)$ to a face of $\Delta(f(x))$. It is clear from the definition of θ that θ takes the simplex $\Delta^{K^n}(x)$ to a face of $\Delta^{K^m}(x)$. Putting these observations together, we see that $s \circ \theta$ takes $\Delta^{K^n}(x)$ to a face of $\Delta(f(x))$. In particular, it takes x to an element of $\Delta(f(x))$.

3. Suppose that $s, t : |K| \rightarrow |L|$ are simplicial maps, and let $s_i, t_i : C_i(K) \rightarrow C_i(L)$ denote the induced maps on oriented i -simplices. Suppose also that, for each i , we have a linear map $d_i : C_i(K) \rightarrow C_{i+1}(L)$ with the property that we have the following equality of linear maps from $C_i(K)$ to $C_i(L)$:

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i.$$

Such a collection of maps d_i is called a **chain homotopy** relating the simplicial maps s and t . Show that we have $t_* = s_* : H_i(K) \rightarrow H_i(L)$.

Solution: Suppose that $z \in Z_i(K)$. We have

$$t_i(z) - s_i(z) = \partial_{i+1}^L \circ d_i(z) \in B_i(L),$$

so $t_i(z)$ and $s_i(z)$ represent the same class in $H_i(L)$. In other words, $t_*([z]) = s_*([z])$.

4. Recall that two simplicial maps $s, t : |K| \rightarrow |L|$ are **close** if, for every simplex $A \in K$, there exists a simplex $B \in L$ such that $s(A)$ and $t(A)$ are both faces of B . The smallest such simplex B is called the **carrier** of A . The goal of this problem is to show that, if s and t are close, then they are related by a chain homotopy, and therefore they induce the same map on homology. Assume that s and t are close.

(i) Define $d_0 : C_0(K) \rightarrow C_1(L)$ by putting

$$d_0(v) = \begin{cases} 0 & \text{if } s(v) = t(v) \\ (s(v), t(v)) & \text{otherwise.} \end{cases}$$

Show that $\partial_1^L \circ d_0 = t_0 - s_0 : C_0(K) \rightarrow C_0(L)$.

Solution: This is immediate from the definition!

(ii) Suppose that $j > 0$, and that, for all $i < j$, we have defined homomorphisms $d_i : C_i(K) \rightarrow C_{i+1}(L)$ such that

$$d_{i-1} \circ \partial_i^K + \partial_{i+1}^L \circ d_i = t_i - s_i$$

and, for any oriented simplex σ , $d_i(\sigma)$ is always a chain in the carrier of σ . Show that, for any oriented j -simplex σ of K , we have

$$\partial_j^L (t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma)) = 0.$$

Solution: By the first equation, we have

$$\partial_j^L (t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma)) = \partial_j^L \circ \partial_{j+1}^L \circ d_j(\sigma) = 0.$$

(iii) Let $M \subset L$ be the subcomplex of L consisting of the carrier of σ and all of its faces. Show that there exists an element $c \in C_{j+1}(M) \subset C_{j+1}(L)$ such that

$$t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma) = \partial_{j+1}^L(c).$$

Solution: First, we observe that $t_j(\sigma) - s_j(\sigma) - d_{j-1} \circ \partial_j^K(\sigma) \in C_j(M) \subset C_j(L)$. In part (ii), we showed that this class is killed by ∂_j^L , so in fact it leaves in $Z_j(M)$. Since M consists of a single simplex and all of its faces, M is a cone. That means that $H_j(M) = 0$, and therefore that $Z_j(M) = B_j(M)$. The result follows.

(iv) Finish the proof that the simplicial maps s and t are related by a chain homotopy.

Solution: We've just proved that, provided that we have a definition of d_i for all $i < j$, we can define d_j with all of the properties we need.